CS771 Mini-Project 1

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Protocol

- **Task:** we predict pollution at 12:00 PM tomorrow (day d+1) using only the data available up to 12:00 PM today (day d).
- **Data split:** training targets are 1 Jan to 31 Aug (the last one is 31 Aug 12:00, which is the first validation issue time). Validation targets are 1 Sep to 30 Nov (91 days $\times$ 468 stations = 42,588 forecasts). Nothing was dropped from validation.
- **Calculations:** averages, scaling and other fitted statistics use only training rows, to avoid cheating. All model choices were made on validation results. We used a fixed random seed (`RANDOM_STATE`= 0), so results can be reproduced.
- **Safety checks:** every lag is checked to be ≥ 0, and a "future-poisoning" test overwrites all readings after the issue time and checks that no feature changes.
- **Models:** only standard non-neural scikit-learn models (Ridge, Lasso, Huber, KNN, random Fourier features, PCA, K-means, Random Forest, boosting). No external data.

Part 0: Cleaning the raw Kanpur data

Format. There are 76 sensors: 44 with 10-digit IDs, 31 with 8-character IDs and one with a 12-character hex ID. The file uses a 15-minute grid, but 75 sensors report only once per hour (at minute :00); the hex sensor reports every 15 minutes. No sensor moves. 10-digit sensors give whole numbers, 8-character sensors give 4-decimal values. Hour by hour, the two main types have very similar medians (about 2 μg/m^3 apart, ratio of means 1.03); the hex sensor reads lower (median 25 vs 33–37).

Hourly data. We made one reading per sensor per hour by averaging, labelled by the hour start. Error codes (like 999) are removed before averaging, so they cannot spoil real values. Averaging vs simply keeping the :00 reading only matters for the hex sensor: the difference is 3.5 μg/m^3 RMS (about 8.5% of its level), with no bias, and the :00 rule would lose 4 hours completely.

Problems found. Bad values are flagged, not deleted. Each threshold is based on a clear pattern in the data:

check	threshold and justification	readings	sensors
error code	value = 999; an isolated spike in the histogram	136	5 (120 from 1204260025)
out of range	$> 500\,\mu$g/m^3; above the sensor range (network 99th pct $\approx$ 108)	28	11
implausible zero	exactly 0; mostly long zero runs of dead sensors	813 h	21 (1203260082: 414)
stuck	≥ 24 identical hours in a row; natural repeats reach 22 h, faulty runs are 74–186 h (clear gap)	929 h	7
mostly unusable	$> 50\%$ of hours flagged; the next-worst sensor is at 15%	600 hours†	2 (1203260082, 1203260068)
network outage	$< 10\%$ of the 76 sensors reporting	117 h	5 Aug 08:00–10 Aug 04:00 UTC
barely follows network	$r < 0.4$ with the leave-one-out network median (median $r = 0.71$)	5,801 h	5

† readings of the two dropped sensors that were not already flagged by another check.

For 65 hours, all 10-digit sensors stopped at the same time. This is not counted as a network outage (the 8-character sensors were still working), but the city average in those hours depends on one sensor type only. The cleaned data is in `kanpur_hourly_clean.csv` (full $76 \times 1{,}488$ hour grid, one `flag` per row); missing hours are left as NaN.

Calibration check. 20 pairs of sensors (a 10-digit and its nearest 8-character sensor, within 3 km) have data in both months. We fitted a straight-line correction on July (8-char $\approx 22.88 + 0.396\times$ 10-digit) and tested it on August:

	RMSE	MAE	mean diff (10d−8c)	corr
July, raw (fit data)	18.64	10.94	−1.38	0.33
July, corrected (in-sample)	16.36	10.68	0.00	0.33
August, raw	20.53	11.92	−1.80	0.57
August, corrected (test)	**21.53**	**13.80**	−1.02	0.57
August, offset-only correction (test)	20.45	12.07	−0.42	0.57

The correction made August worse (RMSE 20.5 → 21.5; only 4 of 20 pairs improved). Why: (1) there is barely any built-in difference between the two sensor types (mean difference only −1.4 to −1.8; subtracting it barely changes anything). (2) Paired sensors up to 3 km apart see different local pollution and do not move together hour by hour (correlation 0.33 in July, 0.57 in August). With such noisy data the fitted slope shrinks to 0.40 (instead of about 1), so the "correction" flattens real pollution changes. True calibration needs sensors on the same pole or a reference monitor.

Summary.

stage	sensors	readings	share missing
raw 15-min file	76	85,762	0.810
hourly table (all values)	76	82,131	0.274
clean, ok only	69	73,824	0.347

In local time (IST), the network's average daily cycle peaks around 06:30 AM (41 μg/m^3) when air is trapped near the ground, and is lowest around 04:30 PM (34 μg/m^3) because of daytime air mixing.

Part 1: Baselines

Persistence predicts that tomorrow's noon pollution equals today's noon reading. Ridge is a basic linear model on 29 features (24 hourly lags, neighbour and network means, sin/cos of the day), trained on 113,256 rows (242 issue days × 468 stations), with negative predictions set to 0.

	overall		Sep		Oct		Nov	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Persistence	32.75	17.47	23.33	8.91	20.22	11.97	47.81	31.70
Ridge baseline	29.58	18.16	20.34	10.35	19.36	13.46	43.06	30.81
Our model	**28.55**	**16.35**	**19.60**	**8.74**	**17.43**	**11.20**	**42.13**	**29.30**

Table 1: validation RMSE and MAE (μg/m^3), overall and per month. Our model is described in Part 2.

No single baseline wins on both metrics. Ridge has the lower RMSE; Persistence has the lower MAE. Training on squared error aims for the average outcome, while MAE rewards the median value. RMSE punishes big mistakes heavily, so Ridge plays safe and predicts typical values: this wins in the tail (99th-percentile error 116 vs 137), which dominates RMSE. But on normal, calm days Persistence is usually closer (Ridge is closer on only 43% of rows), and Ridge's seasonal terms add a +5 μg/m^3 bias in validation.

Part 2: Improving the forecast

Features: 63 features in 8 groups, using up to 7 days of past data.

group	#	content
recent24	24	last 24 hourly readings
daily_hist	6	daily means 2–7 days back
same_hour	7	reading at the target hour (12:00) on each of the 7 previous days
trend	6	today − yesterday, 3-h change, 24-h slope/std/min/max
neighbourhood	8	10-nearest-neighbour mean and means within 10/25/50 km, now and over 24 h
network	4	network mean now, 24 h, 7 days, day-on-day change
station	4	station training mean/std (rows ≤ 31 Aug 12:00), x/y km
calendar	4	sin/cos of target day-of-year and day-of-week

Training rows: 110,448 (236 issue days × 468 stations). The first 7 days are lost because the features need 7 days of history. We use only 12:00 issue times: adding other issue hours (11–13) gave 330,876 rows but did not help (E16).

Experiment log (validation; all models except E3 predict the change in pollution; train RMSE in brackets):

experiment [train RMSE]	RMSE	MAE	experiment [train RMSE]	RMSE	MAE
E1 ridge baseline, 29 feats [28.0]	29.58	18.16	E10 grad. boosting, level [17.1]	35.94	20.12
E2 same, scaled [28.0]	29.58	18.16	E11 grad. boosting, change [17.3]	32.79	18.76
E3 scaled ridge, 63 feats, level [26.3]	29.41	17.88	E12 grad. boosting, shallow [24.0]	30.97	17.15
E4 scaled ridge, α=3000 [26.5]	29.30	17.59	E13 random forest [20.7]	29.94	16.89
E5 lasso α=0.1, 27/63 weights 0 [26.4]	29.27	17.58	E14 ridge + PCA(5) of lags [26.8]	29.41	17.60
E6 Huber loss [26.8]	29.66	16.64	E15 ridge + K-means means [26.5]	29.27	17.55
E7 log-log ridge, smearing [26.9]	30.13	16.69	E16 ridge, issue hours 11–13 [26.8]	29.33	17.68
E8 KNN k=200	30.69	17.13	E18 extra trees alone [22.7]	29.05	16.23
E9 RBF random feats + ridge [25.3]	31.26	19.08	E17 ridge + extra trees, 8 groups [23.9]	28.56	16.42
			E19 final: same, 7 groups [23.9]	**28.55**	**16.35**

Settings were chosen on validation data (ridge α from 1 to 3000, extra-trees leaf size 20/50/100, max features 0.3/0.5). For $\alpha \leq 100$ results are almost the same (< 0.01), so we used $\alpha = 1$. Our final model is the average of a Ridge model and an ExtraTrees model (100 trees, leaf size 50).

Ablation (which groups matter). First we removed groups one at a time from the 8-group model; the trend group made no difference (ΔRMSE -0.001), so we dropped it. Then, as the assignment asks, we repeated this starting from the final 7-group model (same settings). If error goes up when a group is removed, the group helps:

removed	none	recent24	daily_hist	same_hour	neighb.	network	station	calendar
# features in group	57	24	6	7	8	4	4	4
val RMSE	28.555	28.875	28.641	28.636	28.605	28.834	28.570	28.976
ΔRMSE	0	+0.320	+0.086	+0.081	+0.050	+0.279	+0.015	+0.422
ΔMAE	0	+0.077	+0.036	+0.083	+0.058	+0.296	+0.042	+0.032

All 7 groups help, but the small ones (station, neighbourhood) only slightly. Many groups carry similar information (recent24, same_hour and daily_hist all describe recent levels; neighbourhood and network both describe regional pollution, correlation 0.93), so removing one at a time causes only a small drop. The final model beats both baselines on both metrics in every month (-3.5% RMSE vs Ridge).

What worked and what did not

- *Helped most:* calendar (position in the season), then the recent readings and network-level pollution (see the ablation table).
- *Trend features:* just repeat information already in the 24 lags, and today's change says little about tomorrow's, which depends on weather.
- *Log transform:* better MAE but worse RMSE (it under-predicts the very high values that dominate RMSE). *PCA and K-means:* only re-describe information Ridge already has. *Extra issue hours:* more rows but the same days, so little new information.
- *Did a more flexible model beat Ridge?* Boosting and the random forest did not (best 29.94 vs 29.30 for the best Ridge). Extra trees alone did, slightly (RMSE 29.05, MAE 16.23 vs 29.30 and 17.59), but its gap between training and validation RMSE is larger (22.7 vs 29.05, compared with 26.5 vs 29.30 for Ridge).
- *Overfitting:* Gradient Boosting scored 17 on training but 33–36 on validation. It memorised Jan–Aug patterns that do not hold in the Sep–Nov post-monsoon build-up (overfitting under a seasonal shift), and tree models cannot predict beyond values seen in training. Predicting the *change*, and averaging extra trees with Ridge, fixed most of this because the two models make different mistakes.

What missing information would help? Weather forecasts for the target day (wind, rain, boundary-layer height), crop-fire data, festival dates, and knowing which values were gap-filled. The biggest errors were sudden jumps across the whole state (e.g. 29 Nov), which look weather-driven, but we could not test this.

Part 3: Lead time

The target is always 12:00 on d+1; the issue time moves closer. For each lead, Ridge and our model are refitted with all features recomputed relative to the new issue time (same 24-hour window).

		RMSE			MAE		
lead	issue time	Persist.	Ridge	Ours	Persist.	Ridge	Ours
1 h (nowcast)	11:00, d+1	23.95	18.95	**18.86**	10.36	8.67	**7.95**
3 h	09:00	51.43	28.51	27.94	28.28	16.70	14.57
6 h	06:00	66.82	34.06	31.93	41.28	23.16	18.58
12 h	00:00	72.74	31.97	31.04	46.74	21.31	18.28
24 h (day-ahead)	12:00, d	32.75	29.58	**28.55**	17.47	18.16	**16.35**

Table 2: the 1-h nowcast (top) is reported separately from the day-ahead result (bottom) and the intermediate leads.

Persistence: its error rises sharply from 1 h to 12 h (24 $\rightarrow$ 73) and then *drops* at 24 h (33). Pollution has a strong daily cycle: validation averages peak in the evening (about 106 μg/m^3 at 20:00), stay high overnight and fall to about 53 μg/m^3 at noon. A night or morning reading copied onto a noon target is far too high (bias +40 μg/m^3 at 12 h), but at 24 h noon is compared with noon (bias about 0.5). **Learned models** remove most of this because they learn the night-to-noon drop, but they show a mild dip too: 6–12 h are still worse than 24 h, and in November they over-forecast by +9 to +15 μg/m^3 because the needed drop (about -70 μg/m^3) is larger than the drops seen in training. **For public warnings** we would use the 24-hour forecast so people can plan; the cost is about 9.7 μg/m^3 of RMSE compared with the 1-hour nowcast (28.55 vs 18.86).

Part 4: Maps

Station values are interpolated to the 0.05° grid with inverse-distance weighting (power 2, all 468 stations); the

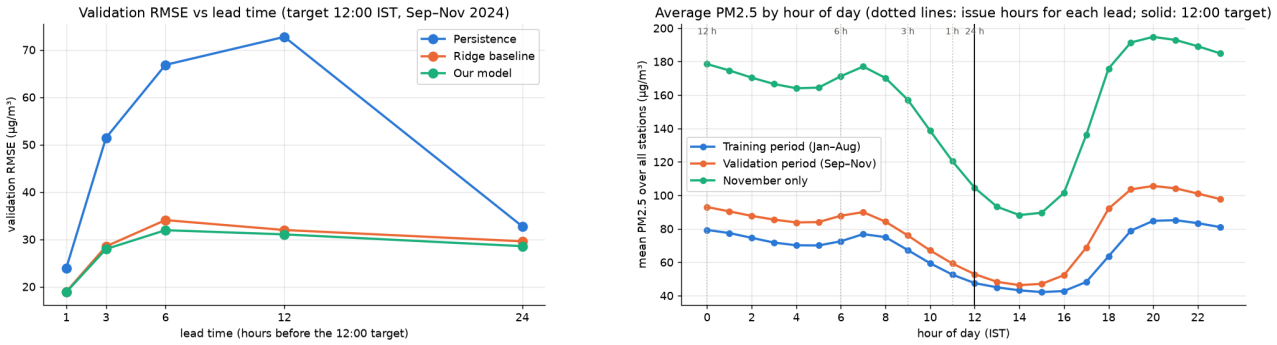


Figure 1: Left: validation RMSE (μg/m^3) vs lead time for Persistence, Ridge and our model. Right: mean PM2.5 (μg/m^3) by hour of day (dotted lines mark the issue hours of each lead; solid line marks the 12:00 target).

same weights are used for observed values, persistence and our forecast. We picked the days by rule: **clean** = lowest observed network mean, **polluted** = highest, **worst** = highest station RMSE of our model.

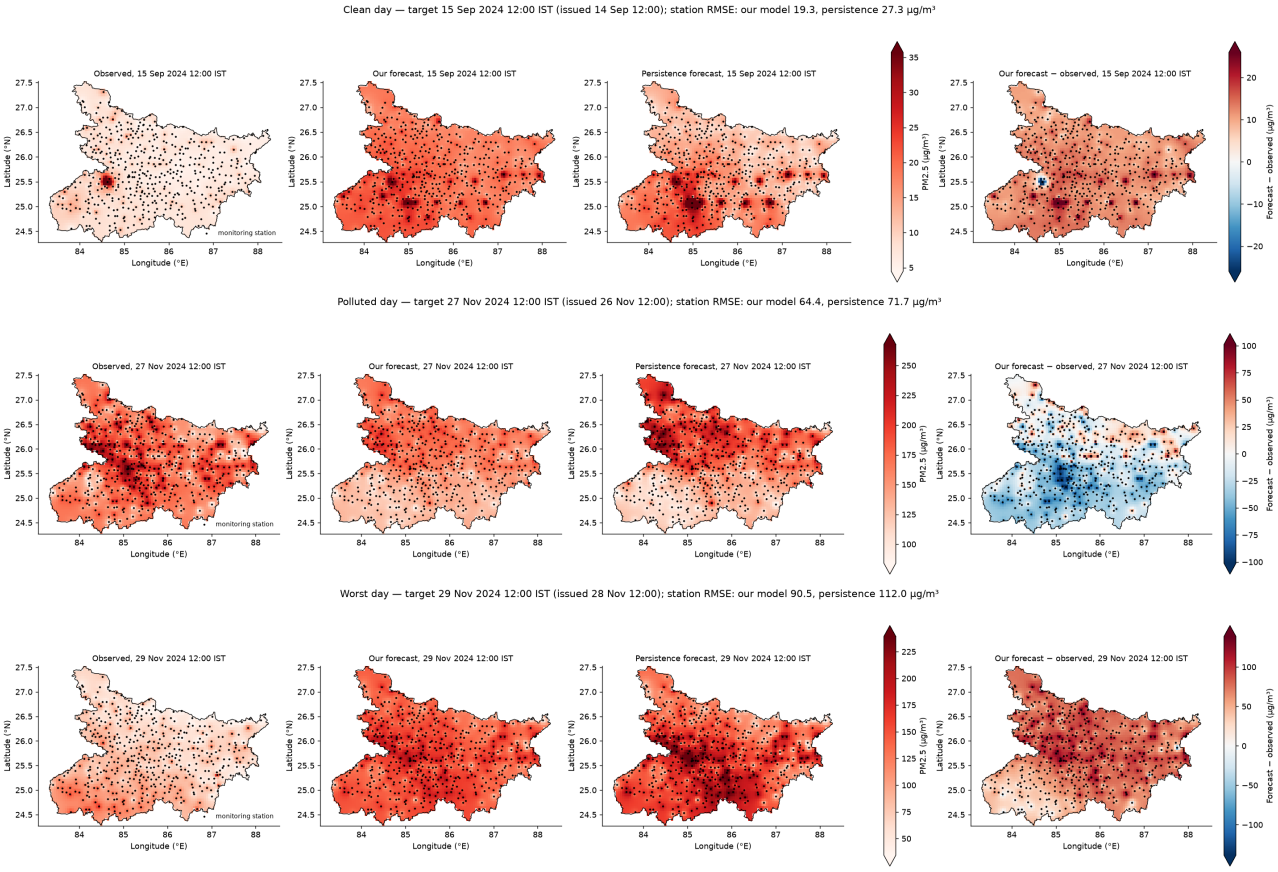


Figure 2: Observed | our forecast | persistence (shared colour scale, μg/m^3) | our forecast − observed. Rows: clean 15 Sep, polluted 27 Nov, worst 29 Nov.

day	obs. mean	station RMSE (ours / pers.)	map RMSE (ours / pers.)	pattern corr. (ours / pers.)	
15 Sep (clean)	7.9	19.3	27.3	13.2 / 13.1	0.58	0.41
27 Nov (polluted)	175.0	64.4	71.7	36.0 / 41.7	0.59	0.51
29 Nov (worst)	76.6	90.5	112.0	76.4 / 86.7	0.38	0.49
26 Nov (extra example)	163.8	51.3	47.7	32.3 / 25.7	0.86	0.85
mean over 91 days	52.8	**24.88**	28.57	**14.17** / 15.37	**0.618**	0.580
pooled station RMSE		28.55	32.75			

Table 3: per-day map metrics for the plotted days (plus 26 Nov, explained below), and averages over all 91 validation days.

On 29 Nov the network average fell from about 159 to 77 μg/m^3 in one day. Both forecasts were too high everywhere (our mean bias is +74 μg/m^3). This is probably a weather change that past PM2.5 data cannot show, but we have not checked the cause.

Which RMSE is better? The map RMSE is lower (14.2 vs 24.9), but it should not be quoted as accuracy. Interpolation smooths both maps (spread drops from 30.4 across stations to 14.0 across grid cells), so noise and extremes cancel and every model looks better. **Station RMSE** compares forecasts with real sensor readings, so we quote that. **Is the observed map the truth?** No. Between stations it is a weighted average of far-away sensors, not a measurement; it cannot show hotspots, never goes above the highest station reading, and carries sensor errors and the unmarked gap-filling. **Good pattern but bad error:** a forecast that is 1.5 times the real values everywhere has perfect pattern correlation but a large RMSE. It happens in our data too: on 26 Nov our pattern correlation is 0.86, but the station RMSE is 51 μg/m^3 (the forecast was too low by 17 μg/m^3 on average). Across all 91 days, pattern correlation and station RMSE are essentially unrelated ($r = -0.03$). For health warnings, getting the *level* right matters more, because warnings depend on crossing health thresholds.

Part 5: Hidden December test

- **Refit:** the final model (same 7 groups, same settings) was retrained on all 12:00 targets up to 30 Nov: 327 issue days, 153,036 rows. Station statistics were recomputed using readings up to 30 Nov 12:00 only.
- **Submission:** `submission/` has `forecast.py`, `model.pkl`, `bihar_stations.csv` and `features.py`. `forecast()` reproduces the notebook predictions to 1.2×10^{-12} μg/m^3 and takes about 0.06 s per call. The two copies of `features.py` (training code and submission) were written slightly differently in three functions; we tested that they give exactly the same features and forecasts.
- **Checker:** `check_submission.py` prints "RESULT: OK".

Limitations

- The data had pre-filled missing values that were not marked.
- Validation covers only Sep–Nov, not December (the worst smog month), so December errors are likely closer to November's (≈ 42) than to the overall 28.6.
- Our gain over Ridge is small (3.5%), probably because we have no weather data for sudden changes.
- The station training mean uses all of Jan–Aug, so training scores are slightly optimistic. Settings were chosen on one validation period, so some selection optimism remains.

Use of AI tools

Claude Code was used for writing and debugging the code (data loading, vectorised feature builders, model experiments, IDW interpolation, `forecast.py`). **claude.ai** was used to generate the LaTeX code for the report and to repair a corrupted notebook file (invalid JSON). Moreover, **Google Gemini** was used to format the report, check for presence of all the answers that are needed and to fix file path errors.